

Sarah Marcum

Planetary Materials | Thermodynamic Modeling | Exoplanet Interiors

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EDUCATION

Ph.D., Geochemistry Expected August 2026

University of California, Los Angeles

Dissertation: *Structure, Phase Stability, and Transport in Hydrogen-Bearing Magma Oceans*

Advisor: Edward Young

M.S., Geochemistry 2026

University of California, Los Angeles

B.S., Physics with Honors 2018

University of Missouri, Columbia

Minors: Mathematics; English Writing

RESEARCH EXPERIENCE

Graduate Researcher 2020–Present

Department of Earth, Planetary, and Space Sciences, University of California, Los Angeles

- Characterized hydrogen-bearing planetary materials using high-pressure/temperature simulations to investigate phase stability and exoplanet interior structure.
- Used high-resolution mass spectrometry of clumped isotopes and statistical models to study formation pathways of abiotically produced methane.

Research Assistant Summer 2018/2019–2020

Astrochemistry Laboratory, NASA Goddard Space Flight Center

Advisor: Dr. Erika Kohler

- Designed and built high-temperature laboratory spectroscopy equipment for measuring exoplanet cloud condensate analogs under controlled experimental conditions.

TECHNICAL SKILLS

Computational Skills: Python, Fortran, density functional theory molecular dynamics simulations (VASP), atomic trajectory analysis, material and transport properties calculations, planet structure modeling, high-performance computing workflows

Laboratory Skills: FTIR spectroscopy, high-temperature vacuum experiments, thermogravimetric analysis, gas chromatography and sample purification, high-resolution isotope mass spectrometry

PUBLICATIONS

Peer-Reviewed Publications

1. Young, E.D., Werlen, A., Marcum, S.P., Stixrude, L., and Dullemond, C.P. (2025). Differentiation, the exception, not the rule: Evidence for full miscibility in sub-Neptune interiors. *The Planetary Science Journal*, 6:251, 15 pp. DOI: 10.3847/PSJ/ae1012

2. Young, E.D., Stixrude, L., Rogers, J.G., Schlichting, H.E., and Marcum, S.P. (2024). Phase equilibria of sub-Neptunes and super-Earths. *The Planetary Science Journal*, 5:268, 16 pp. DOI: 10.3847/PSJ/ad8c40
3. Li, J., Chiu, B.K., Piasecki, A.M., Feng, X., Landis, J.D., Marcum, S.P., Young, E.D., and Leavitt, W.D. (2024). The evolution of multiply substituted isotopologues of methane during microbial aerobic oxidation. *Geochimica et Cosmochimica Acta*, 381, 223–238. DOI: 10.1016/j.gca.2024.06.032

Submitted / In Preparation

1. Marcum, S.P., Stixrude, L., and Young, E.D. *An Equation of State for Supercritical Silicate-Hydrogen Mixtures at sub-Neptune Interior Conditions*.
2. Young, E.D., Marcum, S.P., Werlen, A., and Wulff P.N. *Ice Giants Revisited: Uranus and Neptune as Magma Ocean Worlds*.
3. Marcum, S.P., Stixrude, L., and Young, E.D. *Exoplanet Dynamos: Conductivity of Hydrogen-Bearing Silicate Liquids*.

SELECTED PRESENTATIONS

[†] Oral presentation

- [†] **Goldschmidt**, 2026. *Sub-Neptune Dynamos: Conductivity of Hydrogen-Bearing Silicate Liquids*. Upcoming on 7/17/26.
- [†] **Goldschmidt**, 2025. *An Equation of State for Supercritical H₂–Silicate Mixtures in Sub-Neptunes*. Invited talk.
- [†] **American Geophysical Union Fall Meeting**, 2023. *A Bayesian Approach to Developing Methane as a Biosignature*.
- **Lunar and Planetary Science Conference**, 2022. *Origin of r-Process Radionuclides in the Early Solar System*.
- [†] **NASA Solar System Exploration Science Directors Seminar**, 2020. *In Situ Temperature-Dependent Laboratory Spectroscopy for Exoplanetary Cloud Condensates*. Invited talk.
- [†] **Lunar and Planetary Science Conference**, 2020. *In Situ Temperature-Dependent Laboratory Spectroscopy for Exoplanetary Cloud Condensates*.
- **American Geophysical Union Fall Meeting**, 2019. *Temperature Dependent Spectroscopy for Exoplanet Atmosphere Condensates*.

TEACHING AND OUTREACH

Teaching Assistant

2020–2025

University of California, Los Angeles

EPSS 9: Solar System and Planets (6 quarters)

Science Communications Specialist

Summer 2017/2019–2020

Hubble Space Telescope, NASA Goddard Space Flight Center

- Developed public-facing Hubble Space Telescope observing resources, including the Messier and Caldwell catalogs for amateur astronomers and the viral “What Did Hubble See on Your Birthday?” tool.

- Wrote social media and outreach content translating Hubble science for public audiences.

Telescope Operations Volunteer

2014–2018

Law's Observatory, University of Missouri

HONORS AND AWARDS

- NASA Space Grant Fellowship, 2021
- Distinguished Teaching Assistant Award, 2021
- CRESST II Post-baccalaureate Grant, 2019–2020
- University of Missouri Departmental Honors in Physics, 2018